Grade 4 Science

Unit 2 Assessment — Using Natural Resources for Energy

Name: _____ HR: 4A 4B 4C 4D

Directions: Read each question and choose the ONE BEST ANSWER.

Part 1: Matching (Questions 1–8)

Write the letter of the definition that matches each word.

Vocabulary	Definitions
1. _____ **Generator**	**A.** A gas fossil fuel, mostly methane, found underground.
2. _____ **Nonrenewable Resource**	**B.** A machine that changes motion into electricity.
3. _____ **Solar Cell**	**C.** A resource that can be restored more quickly than it is used up.
4. _____ **Fossil Fuel**	**D.** A fuel formed from the remains of organisms that lived long ago.
5. _____ **Turbine**	**E.** Energy produced by heat that is transferred from inside Earth.
6. _____ **Geothermal Energy**	**F.** A nonrenewable material mined from rock, used for nuclear energy.
7. _____ **Renewable Resource**	**G.** A machine with blades that spin when wind, water, or steam pushes them.
8. _____ **Hydroelectric Power**	**H.** A solid fossil fuel formed from plants in wet, swampy places.
	I. A resource that cannot be restored as quickly as it is used.
	J. A device that converts sunlight to electricity.
	K. Electricity generated by the energy in moving water, usually through a dam.

Part 2: Multiple Choice (Questions 9–25)

Choose the ONE BEST ANSWER for each question.

9. Coal is plentiful right now, but it forms over millions of years. Is coal renewable or nonrenewable?

A. Renewable, because there is a lot of it right now

B. Renewable, because it comes from plants

C. Nonrenewable, because it is found underground

D. Nonrenewable, because it forms much more slowly than it is used

10. Coal, petroleum, and natural gas all form from _____.

A. Rocks that melt deep underground

B. The remains of organisms that lived long ago

C. Wind blowing for millions of years

D. Sunlight stored in the ground

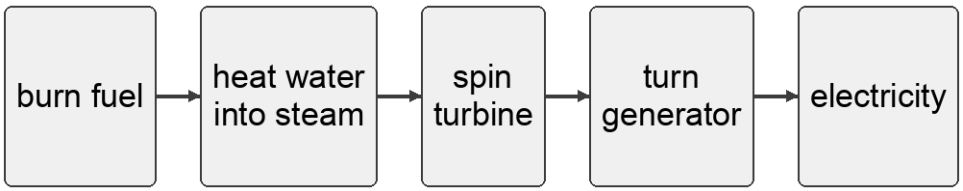
11. Petroleum forms from tiny organisms buried on the ocean floor. Coal forms from _____.

A. Plants buried in swamps

B. Organisms buried in desert sand

C. Rocks buried under mountains

D. Salt deposits under the sea



12. In a power plant, burning fuel heats water into steam. The steam then _____.

A. Turns into coal

B. Turns into petroleum

C. Spins a turbine that turns a generator

D. Cools down and disappears

13. Which fossil fuel is a liquid found under Earth's surface?

A. Coal

B. Petroleum (crude oil)

C. Natural gas

D. Uranium

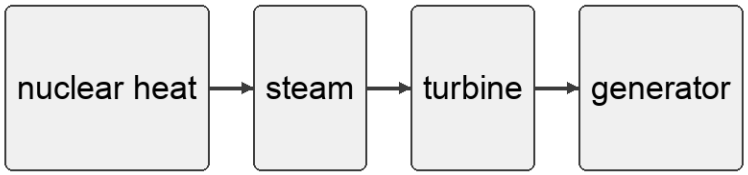
14. Uranium is used to make nuclear energy. Which is true about uranium?

A. It is a renewable resource that grows back quickly

B. It is a fossil fuel made from plants

C. It is a form of moving water

D. It is a nonrenewable material mined from rock



15. A nuclear power plant and a coal power plant are alike because both ____.

A. Burn coal to make heat

B. Use moving water to spin a turbine

C. Use heat to make steam that spins a turbine and turns a generator

D. Use sunlight to make electricity

16. In a wind turbine, what makes the blades spin?

A. Moving air (wind)

B. Sunlight

C. Heat from inside Earth

D. Moving water

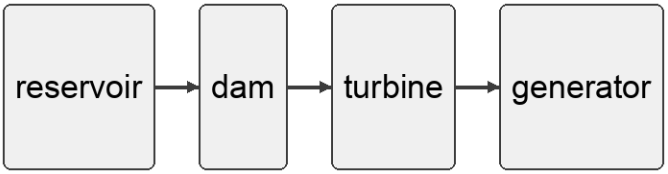
17. Why is wind called renewable but not always available?

A. Wind runs out and can never come back

B. Wind is available every day in every place

C. Wind keeps coming, but it does not blow all the time in every place

D. Wind is made by burning fuel



18. In a hydroelectric dam, what makes the turbine spin?

A. Sunlight hitting the water

B. Moving water flowing through the dam

C. Heat from inside Earth

D. Burning coal under the water

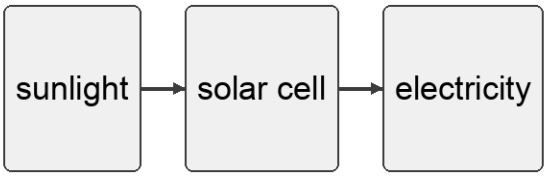
19. Why is hydroelectric power often more predictable than wind power?

A. Water never stops flowing anywhere

B. Dams make their own water

C. Wind is always stronger than water

D. Water can be stored behind a dam and released when needed



20. What does a solar cell do?

A. It converts sunlight to electricity

B. It burns sunlight like fuel

C. It stores wind for later

D. It spins a turbine with steam

21. At night, a solar panel makes little or no electricity. Why?

A. Solar panels only work when they are hot

B. There is no sunlight at night for the solar cell to use

C. Solar panels store too much energy at night

D. Wind blows harder at night

22. Where does geothermal energy come from?

A. Moving air above the ground

B. Sunlight reflecting off the ground

C. Heat transferred from inside Earth

D. Water stored in clouds

23. Why can't every place use geothermal energy?

A. Geothermal works best in places with heat from Earth near the surface, and not every place has that

B. Geothermal energy runs out after one use

C. Geothermal energy is found only in the ocean

D. Geothermal energy needs strong wind all year

24. Regenerative braking turns a car's energy of motion into ____.

A. Gasoline

B. Coal

C. Sunlight

D. Stored energy that can be used again

25. A location has strong sunlight most days but very little moving water. Which resource would work better there, based on availability?

A. Hydroelectric power, because it needs moving water

B. Geothermal energy, because it needs strong sunlight

C. Solar power, because the location has strong sunlight most days

D. Wind power, because it needs strong sunlight